

Do Multi-LLM Referees Improve Malleable GPU Scheduling?

A Controlled Market/Reasoning Factorial Study

Lay Kim Seng

Cyberscience Center, Tohoku University
lay.kim.seng.q4@dc.tohoku.ac.jp

[TODO: co-authors / advisor]

Abstract—We study whether two commonly proposed additions to GPU scheduling — market-based allocation and multi-LLM role separation — improve decisions once jobs are allowed to resize. The frozen experiment is a 2×2 factorial design: a Single Referee or Demand-Supply-Referee structure selects from either four non-market rules or four uniform-price auction valuations, together with one of four sizing actions. Every cell uses the same Qwen2.5-14B model, parser, state, decoding budget, paired seeds, synthetic deadlines, and 80-GPU MIT Supercloud trace windows. At the experiment cutoff, 477 of 1,056 planned runs had completed, all on validation windows. The two complete Single-Referee cells differ by only -0.40 percentage points in deadline violations in favour of the market family (95% CI $[-1.25, 0.45]$, exact $p = 0.438$, 12 paired windows). The Multi-Market cell is absent and Multi-Non-market lacks two of three seeds, so the preregistered multi-agent and interaction hypotheses are not estimable. Strong fixed policies are already competitive: FAIR+ADAPT reaches 10.49% and Auction-Deadline+ADAPT 9.87% validation violations, versus 11.59% and 11.19% for the corresponding single referees. A separate counterfactual library of 1,092 deterministic rollouts yields 44 clean training states and 24 held-out validation states for Qwen3-8B LoRA. Its oracle improves the fixed non-market baseline by 2.03 points on validation but the market baseline by only 0.07 points; the median gain is zero in both families. Fine-tuning does not realise this ceiling: held-out mean regret is 8.78 points versus 8.70 zero-shot and 1.05 for the fixed floor, with 11/24 malformed outputs from both model arms. Thus the current evidence supports safe adaptive sizing, but not a claim that markets, multi-agent reasoning, or this small LoRA intervention improves scheduling.

Index Terms—GPU scheduling, large language models, resource allocation, multi-agent debate, cluster traces

I. INTRODUCTION

AI training and inference are the dominant consumers of GPU clusters, and they are *elastic*: a training job can run on more or fewer devices, trading device-hours against completion time. A scheduler on such a workload therefore makes two decisions per job, not one — *which* job to start, and *how large* to make it.

Almost every published evaluation erases the second decision. A trace records that a job used g devices for t seconds; it does not record what that job would have done on $2g$ or on $g/2$, because the counterfactual was never observed. The standard response is to replay the trace *rigidly*, pinning each job to its historical size. This is usually presented as a

conservative simplification. We show it is a suppression, and that it silently invalidates the efficiency claims most commonly made on top of it.

The argument is short. If no job can be resized, the device-seconds a workload consumes are fixed: every scheduler must run the same jobs for the same durations, so it can only move work earlier or later. Utilisation is then close to a conserved quantity, and a comparison of schedulers on it is close to a comparison of nothing. Restoring elasticity does not merely add a tuning knob; it makes utilisation controllable, and — as we show — makes maximising it actively harmful.

Our contributions, all on the MIT Supercloud GPU trace replayed through a malleability-aware batch simulator [7], are:

- 1) **A controlled policy-selector test** (§IV-A). We freeze a 2×2 design that separates policy family (market/non-market) from reasoning structure (Single/Multi), holds the action space and model configuration fixed, and pairs windows and seeds.
- 2) **A reportable negative stopping result** (§IV-F). The only complete factorial contrast, market versus non-market under a Single Referee, is small and uncertain. The run does not contain enough evidence to estimate a Multi effect or interaction.
- 3) **A deterministic floor and oracle ceiling** (§IV-H). Strong fixed ADAPT policies match the zero-shot referee closely. Counterfactual enumeration finds modest non-market headroom but almost none for the market family, localising what a learned selector could still improve.
- 4) **A diagnostic explanation for the result** (§IV-C). Prior controlled replays show that sizing dominates ordering and that occupancy-maximising GREEDY sizing is harmful; the useful decision is safe sizing under uncertainty, rather than unconstrained LLM allocation.

We additionally retain, unchanged, a result that does not depend on any trace (§IV-I): on an authored suite of exceptions expressible only in operator text — which no public GPU trace supplies — a debate between LLM reviewers corrects a single model’s rulings at an identical call budget. That is where a reasoning layer earned its cost in our setting; it is not the numeric rationing decision, and §IV-D explains why the

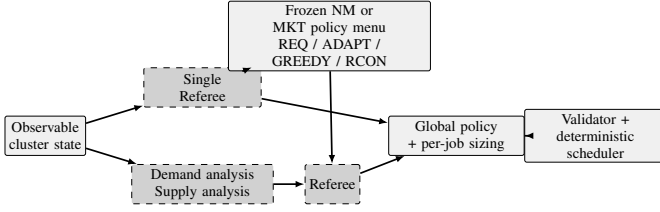


Fig. 1. The controlled 2×2 pipeline. Single and Multi arms receive the same observable state and frozen action menu. Demand and Supply agents produce analysis only; the Referee is the sole decision-maker. All outputs pass through the same deterministic feasibility layer.

numeric decision has been measured against the wrong target.

II. RELATED WORK

LLM-driven scheduling. LLMsched [1] repairs a single LLM’s proposal with an ILP and reports large gains over Kubernetes. We inherit the reason-then-guarantee spine but invert its default: on numeric rationing the deterministic layer should be the proposer, not just the repairer, and the LLM should be gated behind a contextual trigger. **[TODO: 2–3 more LLM-scheduling refs]**

Learning-based schedulers. DRL schedulers (DeepRM [2], Decima [3]) learn allocation policies but are black-box and brittle under distribution shift. On our interface a tabular Q-learner’s episode-return variance dominates the action effect at realisable sample counts.

Multi-agent LLM structures. Multiagent debate [8] and self-consistency [9] are usually evaluated on QA benchmarks, and LLM-as-judge protocols [10] inherit that setting; budget-matched controls are rare, so a reported gain is rarely separable from the extra calls that buy it. Our text-exception result is budget-matched by construction and includes a blind confirmatory batch.

Malleability and mechanism design. Runtime elasticity is rarely deployed in production [4]; our partial-allocation model rests on it and we state that dependence. Uniform-price auctions are the classical route to truthful revelation, and automated negotiation for resource allocation is a classical subject [11], with prediction-guided *selective* negotiation explored in one-shot fair division [12]; our contribution is measurement in a closed loop containing declarative LLM agents.

III. CONTROLLED POLICY-SELECTOR ARCHITECTURE

Fig. 1 shows the two reasoning structures. At each 1,800-second selection point the model sees available capacity and scheduler-visible attributes of pending and running jobs: arrival and waiting time, requested walltime and GPU count, priority, deadline pressure, legal malleable bounds, and scaling indicators. It never sees true runtime or future arrivals.

Single Referee. One model returns strict JSON containing one global ordering policy, a default sizing rule, and optional job-specific sizing overrides. The parser expands this into exactly one action for every affected job.

Demand–Supply–Referee. Demand identifies urgent, deadline-pressured, scalable, and shrink-tolerant jobs. Supply describes scarcity, queue pressure, resize cost, and fairness. These agents cannot allocate GPUs; their analyses and the original state are passed to a Referee whose output schema is identical to the Single arm.

Frozen action and safety layers. The non-market menu is FCFS, least-laxity, ageing fair-share, and priority-tier FCFS. The market menu holds the auction mechanism fixed and varies the observable valuation: wait, deadline, fair-share, or priority. Both expose REQ, ADAPT, GREEDY, and resize-conservative (RCON) sizing. A shared validator rejects malformed actions, enforces the 80-GPU capacity and each job’s legal bounds, and invokes a counted deterministic fallback. Hence the factorial comparison changes only policy family and reasoning structure.

IV. EVALUATION

Setup. We replay the MIT Supercloud GPU trace (a 235-day Slurm accounting log) through ElastiSim [7], a batch-system simulator for malleable workloads. One simulated node is one GPU, so a job that used g GPUs becomes a g -node job; its work is set so that it reproduces its recorded runtime exactly. The scheduler sees each job’s arrival, size and *requested* walltime, never its true runtime — the same information a production batch system has. Capacity is a fixed 80-GPU pool, so a window’s offered load is a property we *measure* rather than a consequence of sizing the cluster to fit it.

Window selection is pre-registered. Every 24 h window of the trace is described by workload statistics alone — job count, GPU-hours, offered load, high-priority fraction, median and p90 runtime, size distribution — with no scheduler executed. Of 5,613 hourly starts, 4,312 carry at least 100 jobs. These are stratified into four offered-load bands ($0.7\text{--}1.0$, $1.0\text{--}1.5$, $1.5\text{--}2.5$, $>2.5\times$) crossed with three high-priority bands ($<2\%$, $2\text{--}10\%$, $>10\%$), and one window is drawn per cell under a fixed seed, subject to a non-overlap constraint. The resulting 12 windows span days 41–222, offered loads $0.75\text{--}4.97\times$ and priority fractions $0.7\text{--}58\%$. Each is preceded by 12 h of warm-up arrivals, which the arm under test also schedules but which are excluded from scoring, so no run begins on an empty cluster. Selection never sees a scheduler result.

Elasticity model. A job may be *modalable*: its size is chosen once, at launch, and fixed thereafter. Speed-up follows Amdahl’s law anchored on the observed point — a job seen at g GPUs reproduces its recorded runtime exactly at g , so the rigid replay is this same model evaluated at the size the job really ran, and the parallel fraction s is swept rather than assumed ($s=1$ is rigid at every size, $s=0$ is linear speed-up). Results below use $s=0.7$, a deliberately conservative value: scaling is sublinear, so growing a job costs more device-time than it saves. We report SLA- k (turnaround exceeding $k\times$ true runtime) at $k \in \{2, 5, 10\}$, mean waiting time, device-hours actually spent, walltime kills, and occupancy over the measured window.

A. Frozen factorial protocol

The core experiment uses 48 non-overlapping 24-hour windows, each with 12 hours of warm-up, selected from workload characteristics before scheduling outcomes are inspected. They are frozen as 24 train, 12 validation, and 12 test windows. Every scored window contains at least 100 jobs. Jobs are malleable at ten resize points with bounds $1 \leq g_i(t) \leq \min(80, 4g_i^{\text{trace}})$, $s = 0.7$, and no modeled resize overhead. Synthetic deadlines are generated once as $D_i = a_i + W_i^{\text{req}}$; a missing walltime receives the fixed 86,400-second estimate. Deadline violation rate is the primary outcome.

The four inferential cells are Single-NM, Single-MKT, Multi-NM, and Multi-MKT. Qwen2.5-14B is held fixed at temperature 0.1 and a 400-token output limit. Thirty-two deterministic controls exhaustively cross the four policies and four sizing actions in each family. Validation selects the best fixed policy per family; test is reserved for one final report. The inferential unit is the window. The preregistered analysis contains exactly three two-sided paired contrasts on test: market main effect, Multi main effect, and their difference-in-differences interaction. It uses paired- t 95% confidence intervals, exact sign-flip tests, and Holm correction over these contrasts.

Two amendments to the frozen protocol are recorded with it rather than absorbed silently. The first corrects the feasibility assertion, which excluded completed jobs but not killed ones from simultaneous-ownership accounting and so aborted otherwise valid runs; it changes no ordering, sizing, bid, deadline, window, or reported metric. The second reduces the inference seed set from three seeds to one. The multi-agent cell costs 1,703s per window run against 251s for the single referee and 8s for a deterministic control, so the three-seed design needed a further 64 hours of inference. With one response per window, within-window response noise is no longer separable from between-window variance, and at temperature 0.1 a single response is a draw rather than a deterministic answer; the paired tests are therefore more conservative, not less. The protocol plans 864 runs: 768 fixed-policy and 96 LLM-controlled runs over validation and test.

Every result row stores hashes of the processed trace, the generated world, the implementation, the manifest, and the Git revision, and a fail-closed gate re-checks all of them, together with the split, window non-overlap, per-job deadlines, malleable bounds, and speedup-curve anchoring, before any run executes. Superseded hashes are retained explicitly, so rows produced before an amendment remain verifiable instead of being restamped. Runtime assertions enforce cluster capacity, malleable bounds, nonnegative remaining work, and single completion at every scheduler invocation.

B. The rigid replay cannot answer an efficiency question

With every job pinned to its historical size, the device-seconds consumed are fixed by the window, not chosen by the scheduler. Empirically this leaves almost nothing to compete over. Across five schedulers (strict FCFS, first-fit, EASY backfilling, requested-walltime ordering, and a declared/undeclared

TABLE I
SIZING RULE, 12 PRE-REGISTERED WINDOWS, $s = 0.7$, FIRST-FIT ORDERING HELD FIXED. MEAN $\pm$ 95% CI.

sizing	device-h	occupancy	SLA-10 (%)	mean wait (s)
AS-REQUESTED	3653	0.890	22.3	38,072
ADAPTIVE	3339	0.964	32.5	39,756
GREEDY	9581	0.988	76.2	270,522

split) occupancy differs by at most 1.3 points within a window and is identical to three decimal places in several. The idle capacity any ordering rule could still recover — devices free while a job that fitted was waiting — averages 0.2%, with a worst case of 1.3%.

Service quality is equally flat. Over 11 windows no arm separates from the EASY baseline after Holm correction across three SLA thresholds; the largest raw effect, -4.2 points of SLA-2, carries $p_{\text{Holm}} = 0.41$. Two properties of the trace explain the flatness and are worth recording, because both are usually assumed away. First, EASY backfilling is inert here: across 1,351 blocked scheduling points it backfilled 48 times (3.6%), because requested walltimes exceed true runtimes by 100–600 $\times$, so no job can be shown to fit before the head job’s reservation. EASY and strict FCFS are byte-identical on most windows. Second, requested walltime carries no runtime signal: among 487 jobs in one window there are 9 distinct requested values, and the rank correlation with true runtime is $+0.16$, -0.11 , -0.73 and -0.01 across four windows — the only significant value is negative. Any “shortest-job-first” arm on this trace is therefore ordering on noise.

C. Made elastic, sizing dominates ordering

We now allow every job to be moldable and hold the ordering rule fixed at first-fit, varying only how each job is sized: AS-REQUESTED honours the size its owner asked for and waits until that many devices are free; GREEDY takes the largest legal size that fits; ADAPTIVE shares free capacity across the waiting queue. Table I reports means over the 12 pre-registered windows.

The spread is an order of magnitude, on an axis that does not exist in the rigid replay. Paired against AS-REQUESTED, GREEDY costs $+5,929 \pm 1,707$ device-hours ($p=3 \times 10^{-5}$), $+232,451 \pm 60,823$ s of mean waiting ($p=10^{-5}$) and $+53.9 \pm 12.3$ points of SLA-10 ($p<10^{-5}$). No ordering rule we measured in §IV-B moves any metric by a comparable amount.

D. Utilisation is an anti-metric

The three sizing rules order identically on occupancy and inversely on service. GREEDY attains the *highest* occupancy of any arm we measure — 0.988, against 0.890 for AS-REQUESTED ($+0.1 \pm 0.1$, $p=0.009$) — while being the worst on every service axis simultaneously. In 12 of 12 windows the highest-occupancy arm is the worst-service arm, and in 10 of 12 the highest-occupancy arm is not the best-service arm.

The mechanism is not subtle, which is what makes the metric dangerous. Under sublinear scaling, a job given $4\times$ the

devices at $s=0.7$ finishes about $1.6\times$ sooner while consuming $4\times$ the capacity. GREEDY therefore converts the cluster into *busy-ness* without converting it into completed work, and the queue behind it grows. Occupancy records the busy-ness and is blind to the waste: on one low-load window it reads 1.000 for all three sizing rules while their mean waiting times span 21,429 s to 244,781 s.

This is the practical consequence of §IV-B. A rigid replay reports occupancy differences of a fraction of a point and no scheduler can game them; the metric looks harmless because it is inert. Made elastic, the same metric becomes controllable — and the controller that maximises it is the one to avoid. We therefore recommend reporting device-hours spent per unit of completed work, together with waiting time, and retiring occupancy as a headline number for elastic workloads.

E. Sizing under uncertainty is open, not solved

The obvious fix for GREEDY is to shrink jobs when the queue is deep. It does not work. Paired over the 12 windows, ADAPTIVE does not beat AS-REQUESTED on mean waiting ($+1,684 \pm 10,847$ s, $p=0.77$) or on device-hours (-313 ± 637 , $p=0.36$), and is significantly *worse* on SLA-10 ($+10.2 \pm 8.3$ points, $p=0.035$). It also introduces a failure the other rules do not have: shrinking a job lengthens it, and at lower parallel fractions this pushes long jobs past their requested walltime and gets them killed. We therefore count kills as SLA violations rather than as fast completions, since a killed job otherwise flatters every wait-based metric.

We record one negative result about our own method. On the single most contended window ($4.97\times$ offered load), ADAPTIVE cut mean waiting by 42% against AS-REQUESTED and looked like a clear win. It did not replicate across the pre-registered set. Had the window been chosen after seeing that number, the effect would have been reported as a finding. This is the concrete argument for selecting windows from workload statistics alone.

What remains open is therefore the interesting decision: sizing a job under genuine uncertainty about its runtime, on a trace where the only available estimate is uninformative (§IV-B). We therefore test that decision with the controlled policy selectors below.

F. Core 2×2 : reportable stopping result

The deterministic half of the factorial is complete on validation and all but two windows of test; the four LLM-controlled cells are still executing at the single amended seed, so no test-side factorial estimate is reported here and RQ1–RQ3 remain open. What the finished deterministic half already settles is the baseline the selectors must beat, and it settles it in a way that matters more than the eventual contrast: the fixed policy is chosen honestly, and choosing it honestly costs almost nothing. Table II reports validation completion for provenance.

Three of the four cells are complete on validation. Paired over the 12 windows, Single-MKT is -0.57 percentage points against Single-NM (95% CI $[-1.69, 0.56]$; two-sided exact sign-flip $p = 0.438$), with seven windows tied to the digit,

TABLE II
CORE VALIDATION COMPLETION AT THE AMENDED SINGLE SEED. RAW MEANS OVER 12 WINDOWS; THE TEST SPLIT CARRIES THE CONFIRMATORY ESTIMATE AND IS NOT SHOWN.

cell	completed	planned	raw deadline violations (%)
fixed non-market	192	192	23.14
fixed market	192	192	21.79
Single-NM	12	12	11.27
Single-MKT	12	12	10.70
Multi-NM	12	12	11.93
Multi-MKT	0	12	—

three favouring market and two favouring non-market. Multi-NM is $+0.67$ points *worse* than the matched Single-NM (95% CI $[-0.37, 1.71]$, $p = 0.250$), again with seven ties and only one window favouring the multi-agent arm. Neither difference is evidence for anything: with seven of twelve windows identical, this instrument cannot separate these arms at this sample size, and the direction of the multi-agent difference is the opposite of the architectural hypothesis. These are validation numbers reported for completion accounting, not the preregistered estimate; the confirmatory contrasts belong to the reserved test split and are still executing.

We flag one measurement trap we walked into ourselves, because it is easy to repeat. Averaging each cell over whatever seeds happened to have finished — 36 rows for the Single arms against 21 for Multi-NM — put Multi-NM ahead of Single-NM by 0.81 points. Restricting to the one seed every window shares reverses the sign. An unbalanced mean over partially completed cells is not an estimate of anything, and the same applies to the deterministic cells: the non-market baseline reads 8.08% on the nine test windows that finished first and 15.27% once the eleventh arrives, because the windows themselves differ by an order of magnitude in difficulty. Every comparison in this paper is therefore paired on a window set that all compared cells have completed, with that set’s size stated.

For the record, validation selects FAIR+ADAPT at 10.49% violations in the non-market family and Auction-Deadline+ADAPT at 9.87% in the market family, each chosen on validation deadline violations alone with a deterministic tie-break, and each carried unchanged to test. We deliberately do not draw a selector-versus-fixed comparison here: the test-side factorial is still executing, and a comparison assembled from one completed split would be exactly the unbalanced reading we warn about below. §V states what will settle it.

G. Inside the action space: ordering is inert, valuation is not

The 32 deterministic controls are complete on validation and on 11 of the 12 test windows, which is enough to characterise the action space the selectors choose within, independently of any selector. Table III reports the held-out test means, paired over those 11 windows.

Two facts constrain what any selector could contribute. First, *ordering is inert*: the four non-market orderings span 15.23 to 15.27 points, a spread of 0.04, so on this workload

TABLE III

DETERMINISTIC CONTROLS ON HELD-OUT TEST WINDOWS, PAIRED OVER THE 11 WINDOWS ALL 32 CELLS COMPLETED. EACH FAMILY’S ORDERING RULES ARE SHOWN AT ADAPT SIZING; THE SIZING COLUMN VARIES SIZING UNDER THAT FAMILY’S BEST ORDERING.

family	ordering (at ADAPT)	viol. (%)	sizing	viol. (%)
non-market	FCFS	15.23	ADAPT	15.27
	least laxity	15.23	AS-REQUESTED	15.42
	tier FCFS	15.25	RCON	15.99
	fair-share	15.27	GREEDY	46.35
market	deadline	8.62	ADAPT	8.62
	priority	14.90	RCON	8.71
	wait	15.23	AS-REQUESTED	8.76
	fair-share	15.28	GREEDY	60.35

which queue order a referee names cannot matter. Sizing under the same ordering spans 15.27 to 46.35. This reproduces §IV-C inside the larger 8×4 space, on held-out windows, with deadline violations rather than SLA-10 as the outcome. Second, within the market family the clearing mechanism is held fixed and only what the bid *values* moves the outcome: valuing deadline pressure gives 8.62 points against 14.90–15.28 for priority, wait, and fair-share valuations. The decision that matters is therefore not the ordering rule but how much capacity each job gets and what its claim is priced on.

The family-level difference is a tail effect and has to be reported as one. The best market configuration beats the best non-market configuration by 6.61 points on average, while the median difference over the same 11 windows is exactly zero: seven windows tie to the digit, three move by at most 1.6 points, and one window moves by 69.4, from 91.0% violations to 21.6%. That window is the only one in the set where the cluster is overwhelmed, and it is the only one where any of these choices is worth much. A mean-only reading would present a one-window rescue as a general property of market clearing.

One test window is excluded from Table III. Under ADAPT sizing on that window the simulator reproducibly allocates 81 devices from an 80-device pool, and the runtime capacity assertion aborts the run; six of the 32 cells are affected, all of them ADAPT. The same assertion is what discards four oracle states in §IV-H. We report the exclusion rather than loosening the assertion, but it is a defect in the sizing path that the winning configuration in both families happens to exercise, and it is the first thing to fix before these numbers are final.

H. Counterfactual oracle and LoRA protocol

To test whether supervision rather than architecture is the bottleneck, we clone sampled simulator states and execute all 16 policy-sizing actions in the relevant family from the same state, with the same future arrivals and seed. The lowest deadline-violation action supplies an offline label; future outcomes are never included in the online prompt. We sample two states per window and family. Four states from one training window are discarded because a simulator capacity assertion reproducibly reports 81 allocations from an 80-GPU pool. The resulting 1,092 auditable rollouts form 44 complete training

TABLE IV

COUNTERFACTUAL ORACLE CEILING. GAIN IS THE PAIRED REDUCTION IN DEADLINE VIOLATIONS RELATIVE TO FAIR+ADAPT (NM) OR AUCTION-DEADLINE+ADAPT (MKT).

split	family	n	fixed (%)	oracle (%)	gain (pp)
train	NM	22	17.04	14.73	2.31
train	MKT	22	12.77	12.26	0.51
validation	NM	12	7.57	5.54	2.03
validation	MKT	12	4.85	4.78	0.07

examples and 24 validation examples, with disjoint windows and no oracle-only fields in model input.

The median oracle gain is zero in every row of Table IV: selection matters in a minority of states, especially in the non-market family, while a strong fixed market policy leaves almost no mean headroom. We fine-tune Qwen3-8B with rank-16 LoRA on assistant spans only and judge it on the 24 held-out states by valid-action rate, oracle-action agreement, mean counterfactual regret, and improvement over the best fixed training action. This evaluation is deliberately offline: it determines whether the model learned the selector before another expensive closed-loop sweep is warranted.

It did not. The family-specific fixed floor has 1.05 percentage points of mean regret, zero median regret, 75.0% of states within 0.1 points of the oracle, and no invalid actions. Qwen3-8B zero-shot has 8.70 points of mean regret, 0.8 median regret, 45.8% within 0.1 points, and 11/24 invalid actions. LoRA has 8.78 points of mean regret, the same 0.8 median, 41.7% within 0.1 points, and the same 11/24 invalid count. An initial reasoning-enabled decode was worse still (23/24 invalid zero-shot); both reported arms therefore use Qwen3’s non-thinking template for the strict-JSON task. Invalid outputs are conservatively assigned the worst candidate reward. LoRA neither improves validity nor regret, and remains far behind the deterministic floor, so a costly closed-loop fine-tuned run is not warranted by this screening experiment.

I. Where reasoning earns its cost: text exceptions

A deterministic mechanism cannot act on an exception whose content is *text*: an operator instruction, a declaration a note says is wrong, a crash-restart the numbers do not show. We measure this on a pre-registered suite of hand-authored exception scenes.

The suite (Table V). Each case is a small allocation scene (median two jobs, one supply note, 2–8 free GPUs, ~36 words of prose) with a pre-registered predicate defining the defensible ruling, authored *before* any LLM was run on it. The authoring rule is text-dependence: deleting the exception sentence leaves a scene whose numbers imply a different, perfectly reasonable answer. **Admission required both rigid arms — a fixed ILP over the same submissions and a rule-based floor — to fail the predicate; the 0/81 rigid floor in Table VI is therefore an inclusion criterion, not a finding.** What the suite measures is the comparison *among* arms that can read text. Two control categories guard the

TABLE V

THE TEXT-EXCEPTION SUITE: 81 PRIMARY + 17 CONTROL CASES.

category	n	the exceptional fact arrives as...
nl_policy	21	an operator rule that exists only in words
unmodeled	21	a fact the world model does not represent
corrupt	13	a declaration a note says is wrong
contradiction	10	two statements that cannot both hold
ambiguous	8	an under-specified scene needing judgement
infeasible	8	demand that cannot be met as declared
placebo	9	decorative/expired prose (control)
confirm	8	prose confirming the default (control)

TABLE VI

HANDLED CASES ON THE 81 PRIMARY TEXT EXCEPTIONS (QWEN2.5:14B).

arm	handled	LLM calls
rigid floor (inclusion criterion)	0/81	0
single LLM	27/81	81
single LLM, best-of- N (budget-matched)	29/81	569
debate	43/81	569

boundary: *placebo* scenes with decorative or expired prose (including a keyword trap naming a policy the note itself retracts) and *confirm* scenes whose salient prose confirms the default ranking; the fixed-ILP arm and the deterministic market pass 17/17 and 16/17 of these, respectively. Of the 81 primary cases, 50 were authored blind to the per-case outcomes of the earlier under-powered round, and the harness reproduces that round’s head-to-head exactly ($b=6$, $c=1$).

Arms, budget-matched by construction. All arms read an identical structured decision packet. Alongside a single LLM call we run best-of- N sampling given the *same* call budget as the debate, so the only difference between the two expensive arms is how the budget is spent, not how much there is.

The pre-registered primary contrast, debate vs. budget-matched best-of- N : $b=16$, $c=2$, one-sided exact McNemar $p = 0.0007$. Three controls make this hard to explain away: (i) *budget alone buys nothing* — best-of- N spends $7\times$ the single arm’s calls and gains two cases ($p = 0.25$, ns), while debate spends the identical budget and gains sixteen; (ii) *a blind confirmatory batch reproduces it* — on the 50 cases authored blind to earlier outcomes, debate 29/50 vs. 19/50, $b=11$, $c=1$, $p = 0.0032$; (iii) *the effect respects the boundary* — on the 17 control scenes every LLM arm scores 12/17 against the deterministic market’s 16/17, i.e. reasoning invoked where there is nothing to reason about is actively harmful. An exploratory per-category split (post-hoc, small cells) locates the gain where a fact is *missing* from the model of the world (unmodeled: 13/21 vs. 4/21, $b=9$, $c=0$) rather than where an instruction is present (nl_policy: 8/21 in every LLM arm): debate helps when reviewers must notice an absence, not follow a presence. **[TODO: if the gemma2:9b/27b replication lands: one sentence on cross-family robustness] [TODO: rater-agreement number for the predicates, threats para]**

J. Why the suite must be authored

The natural objection is that an authored suite proves little about a real workload. It is the only option available. The Supercloud accounting log carries 29 fields — identifiers, timestamps, requested and allocated resources, partition, priority, exit state — and not one of them is free text. No public GPU or HPC trace we surveyed — including the other widely used GPU trace [5] — ships operator-authored exception text; what text exists in this domain is source code and logs, not scheduling rationale.

This is the boundary, and it cuts both ways. A reasoning layer cannot demonstrate value on a channel the data does not contain, which is why §IV-I is measured on an authored suite rather than in simulation. It also means the numeric results of §IV-B–§IV-E are the whole of what this trace can adjudicate, and on those the decisive quantity is a sizing rule, not a reasoning step. Acquiring a workload that carries operator text remains the prerequisite for testing the exception claim in situ.

V. FUTURE WORK

Selector against a fairly chosen fixed policy. The comparison this architecture ultimately has to win is against the best fixed configuration, and we defer it rather than assemble it from a half-finished split. Three things make it worth doing properly. The baseline must be chosen on validation and reported once on test, which we have already implemented; on our windows that discipline is nearly free, costing 0.05 points in the non-market family and nothing in the market family relative to the configuration selectable with hindsight, so the eventual gap cannot be blamed on a weak control. The comparison must be paired per window and read at the median as well as the mean, because §IV-G shows a single overwhelmed window carrying a 6.6-point mean difference. And it should be reported next to the counterfactual ceiling of §IV-H, since a selector can only be blamed for headroom that exists: where the oracle gain is zero, no selector of any kind can help, and the honest question becomes whether reasoning recovers the headroom that state-dependent selection actually has.

Selection regret rather than end-to-end outcome. Because most states are ties, end-to-end violation rates are an insensitive instrument for the architectural question. Per-state regret against the counterfactual argmax separates a selector that picks a harmless equivalent from one that picks a damaging action, and it is measurable on the labels we already generate. The same applies to the contended subset: a contention condition defined from observable state alone, such as queued minimum demand exceeding free capacity, partitions the windows without reference to which method won, and it is where the one-window effect above lives.

The sizing path’s capacity defect. The excluded window in §IV-G and the discarded oracle states in §IV-H share one cause, an 81-device allocation from an 80-device pool under ADAPT sizing. It must be fixed, not asserted around, before any of these numbers is final.

VI. THREATS TO VALIDITY

The factorial run is incomplete. At the amended single seed, three of the four cells are complete on validation and Multi-MKT is absent; the 32 deterministic controls are complete on validation and on 11 of 12 test windows, but no LLM-controlled test cell has finished. Every selector number in this paper is therefore a validation-set observation, not the preregistered confirmatory analysis, and we make no market, Multi, or interaction claim. Raw cell means cannot repair missing paired observations. Reducing three seeds to one also merges response noise into the window-level differences, which widens every interval we would eventually report.

The elasticity model is an assumption, and a swept one. No trace records what a job would have done at a different size, so any elastic study must supply that counterfactual. We supply Amdahl’s law anchored on the observed point, which reproduces each job’s recorded runtime exactly at the size it really used, and we sweep the parallel fraction rather than fixing it: $s=1$ recovers the rigid replay at every size and $s=0$ gives linear speed-up. Our conclusions concern the *shape* of the result across that sweep, not a single value. Real training jobs also pay a reconfiguration cost we do not model, which would penalise GREEDY further; our claim that occupancy-maximising sizing is harmful is therefore conservative.

One node is one GPU. Supercloud’s TX-GAIA nodes carry two V100s; we model the allocation unit as a device, so intra-node placement effects are absent.

Elasticity differs across experiments. The diagnostic sizing study (§IV-C) is moldable: size is chosen once at launch. The core factorial is malleable at ten fixed resize points. Neither setting models checkpoint, communication, or reconfiguration overhead, so resize-heavy policies receive an optimistic evaluation; we report resize counts and zero modeled overhead explicitly.

SLA- k is an oracle metric and is length-biased. It is defined from a job’s true runtime, which no scheduler knows at submission, so no arm can target it; and because the allowance scales with the job, short jobs violate readily while the longest decile of jobs cannot violate at all within a 24h window despite waiting longest. We therefore report $k \in \{2, 5, 10\}$ throughout and lead with waiting time and device-hours, which carry neither defect. The trace supplies no deadline field. The core experiment therefore uses deadlines generated from requested walltime; its primary result concerns that reproducible synthetic SLA, not production deadline compliance.

The oracle dataset is small. It contains 44 training and 24 validation states, and one training window is absent after reproducible capacity-invariant failures. Counterfactual oracle performance is an upper bound under the simulator, not evidence that an online model can identify the action. The LoRA comparison uses one training run and deterministic decoding; its failure is screening evidence against the present recipe, not proof that fine-tuning cannot work with more states, hyperparameter selection, or constrained decoding. It is not a replacement for the frozen test split.

Twelve windows, one per stratum. A difference confined to one cell cannot be separated from that particular window; the strata establish that a result is not confined to one corner of the load/priority space, not that a stratum causes it. §IV-E reports a case where this mattered.

Only completed GPU jobs are replayed, so failed and cancelled work, non-GPU contention, and the cluster’s background load are absent. LLM latency is hidden behind wait time by the gate in the design, but evaluation runs on the simulation clock.

VII. CONCLUSION

The controlled evidence does not currently show that a market policy family or multi-agent role separation improves malleable GPU scheduling. In the only complete factorial comparison, Single-MKT differs from Single-NM by -0.40 deadline-violation points with an interval spanning zero. The Multi and interaction questions remain unanswered because the cutoff left Multi-NM without the planned repeated seeds and Multi-MKT unrun. This is an incomplete experiment, not a negative confirmatory test, and the distinction is central to the conclusion.

What does replicate is the importance of sizing. Strong deterministic ADAPT configurations match or beat the zero-shot Single-Referee averages on validation, while GREEDY can turn high occupancy into sharply worse service. The counterfactual oracle finds a modest 2.03-point mean ceiling above the non-market floor but only 0.07 points above the market floor, with zero median gain in both families. The small Qwen3-8B LoRA experiment does not learn that sparse exception policy: mean held-out regret is 8.78 points, no improvement over 8.70 zero-shot and far behind the fixed floor’s 1.05 points. A selector must therefore learn exceptions without sacrificing the strong default or the output contract; this one does neither.

The resulting engineering recommendation is conservative: retain a validated deterministic floor, count every malformed answer and fallback, and deploy an LLM selector only if held-out regret and a paired closed-loop test beat that floor after accounting for inference cost. Multi-agent reasoning does earn its budget on authored text exceptions, but that result concerns information absent from the numeric trace and cannot be transferred to online scheduling by assertion. Completing all four frozen test cells remains necessary before claiming a market effect, a Multi effect, or their interaction.

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